

Roll No.....

ODD END Semester Examinations 2025

Name of the Program: **B.Tech-ECE**

Name of the Course: **Electronics Engineering Materials**

Time: **3 Hours**

Semester: **III**

Course Code: **TEC 343**

Maximum Marks: **100**

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub-questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	What are line defects in solids?	CO1, CO3
(b)	State different types of matter and define grain boundary.	
(c)	Explain direct and inverse piezoelectric effect and define electromechanical coupling factor.	
Q2	(20 marks)	
(a)	Calculate the wavelength of an electron with kinetic energy of 300 eV. Given charge of an electron = $1.6 \times 10^{-19} \text{ C}$, mass of an electron = $9.1 \times 10^{-31} \text{ kg}$, $h = 6.626 \times 10^{-34} \text{ Js}$.	CO2, CO3
(b)	A conducting wire has a resistivity of $1.5 \times 10^{-8} \text{ ohm-m}$ at 300 K. Its fermi energy is 5 eV. Its volume of 1 m^3 contains 6×10^{28} conducting electrons. Hence obtain the value of drift velocity of electrons, when the conductor is subjected an electric field of 100 V/m. Given, charge of an electron = $1.6 \times 10^{-19} \text{ C}$, mass of an electron = $9.1 \times 10^{-31} \text{ kg}$.	
(c)	The number of atoms in hydrogen gas is $9.8 \times 10^{20} \text{ atoms/cc}$. The radius of hydrogen atom is 0.053 nm. Calculate its electronic polarizability and relative permittivity.	
Q3	(20 marks)	
(a)	Explain polarization and state electric displacement vector in terms of dielectric susceptibility.	CO3, CO5
(b)	State fundamental absorption and exciton absorption in solids.	
(c)	Distinguish Type 1 and type 2 superconductors.	

Q4	(20 marks)	CO3, CO4
(a)	Explain free carrier absorption and impurity absorption.	
(b)	Explain electronic polarization and derive its expression.	
(c)	A magnetic material has a magnetization of 2300 A/m and produces a flux density of 0.00314 Wb/m ² . Calculate magnetizing force and relative permeability of the material.	
Q5	(20 marks)	CO5, CO6
(a)	Elaborate on types of resistors and its components.	
(b)	Explain thermoplastic and thermosetting plastics with example.	
(c)	The transition temperature for Pb is 7.2. However, at 5 K it loses its superconductivity property if subjected to magnetic field of 3.3×10^4 A/m. Find the maximum value of H which will allow the metal to retain its superconductivity at 0 K.	